

Oleksii Krasnoshtanov

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English (fluent) | Ukrainian (native) | Russian (fluent) | Available from February 2027

EDUCATION

Breda University of Applied Sciences

BSc Applied Data Science & Artificial Intelligence; 9.1/10 (4.0 GPA); BUas Scholarship

Breda, Netherlands

Sep 2024 – Present

PROJECTS & TECHNICAL WORK

MARBET Event Assistant – On-Premise RAG Chatbot | *2-Person Sprint* | [Code](#)

Apr – May 2025

- Self-hosted RAG assistant answering event questions from the client's own confidential documents, built in a 2-person sprint and deployed to their production: Llama 3.3 70B (q5_K_M) served on-premise through Ollama
- Retrieval built for answer quality: semantic chunking on paragraph then sentence boundaries (800/200) over fixed cuts, mx-bai-embed-large embeddings in FAISS, two-stage fetch-then-filter (fetch_k 20, top-k 8)
- Grounded by construction: answers only from retrieved context, each returned with the source chunks and their relevance, and an explicit fallback when the documents do not cover the question

CV Pipeline – Plant Phenotyping Platform | *5-person group project, BUas* | [Code](#) | [Docs](#)

Apr – Jun 2026

- Built an MLOps platform for plant organ segmentation and root-tip detection: U-Net (ResNet-34) behind an authenticated FastAPI backend, with a Next.js frontend, PostgreSQL and Prometheus
- Ran the model lifecycle from Airflow DAGs: data versioning, Azure ML training, drift detection on confidence, latency and error rate, and retraining driven by researcher feedback on flagged predictions
- Owned deployment and CI/CD: one codebase shipping to Docker Compose, on-premise Portainer and Azure Container Apps, with blue-green cutover, health-gated rollback and OIDC cloud auth; 85% coverage gate, 430 tests

Emotion Analysis Pipeline | *Group Project – Content Intelligence Agency*

Sep – Nov 2025

- Generated synthetic training data with Mistral to fill the gaps in a merged multilingual corpus, then fine-tuned DeBERTa and Russian transformer models on it, cross-validating across languages to stabilise predictions
- Production NLP pipeline for a Dutch media client, processing 40-minute episodes end-to-end: Faster-Whisper V3 transcription with VAD, multilingual emotion classification and automated visualisation

Dutch Sign Language Fingerspelling Trainer | *Solo rebuild of a group project* | [Code](#) | [Docs](#)

May – Sep 2026

- Real-time NGT alphabet trainer: MediaPipe landmarks feed a PyTorch residual MLP for the 24 static letters (95% validation) and a bidirectional LSTM over 30-frame sequences for the two needing motion (92%)
- Sole author: FastAPI WebSocket backend streaming to a Next.js 16 / TypeScript frontend, bilingual EN/NL, containerised with CI and published documentation

NPEC Plant Phenotyping System | *Individual Project, BUas* | [Code](#)

Nov 2025 – Jan 2026

- Ranked **#2 of 77** in a Kaggle root-segmentation competition and selected to present; built the full loop from U-Net segmentation and root-tip extraction to OT-2 pipette control in PyBullet, PID benchmarked against PPO

HACKATHONS & COMPETITIONS

BrabantHack 2026 | *Winner – Demcon Deep Tech Track* | [Code](#)

Apr 2026

- Won the Demcon track with a model that locates a pedestrian standing entirely outside the camera frame from the shadow they cast into it: 0.626 mean IoU on the organisers' hidden test set; team of 3
- Drove the gain by re-describing the box relative to the frame edge it hides behind – side, distance, width, height, y-centre – on a ResNet-50 trunk fused with 19 shadow descriptors: 0.468 to 0.610 mean IoU, reproducible from the repository

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript

Generative AI & LLMs: LangChain, Ollama, FAISS, Llama 3.3, RAG, semantic chunking, prompt grounding, Gradio

ML / Deep Learning: PyTorch, Scikit-learn, XGBoost, HuggingFace Transformers, Faster-Whisper

Computer Vision: OpenCV, U-Net, MediaPipe, PyBullet

MLOps & Cloud: Azure ML, Azure Container Apps, Airflow, MLflow, Prometheus

Infrastructure: Docker, Docker Compose, Portainer, GitHub Actions CI/CD, uv, Ruff, pytest

Backend & Data: FastAPI, Pydantic, PostgreSQL, SQLAlchemy, Alembic, Pandas, NumPy

Frontend: Next.js, React, Tailwind CSS, WebSockets